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ClimateScope

Milestone 1: Data Preprocessing and Initial Analysis Report

1. Introduction

The purpose of this milestone is to ensure the raw weather dataset is transformed into a structured, clean, and analysis-ready format. This step is crucial for building reliable models, maintaining data integrity, and supporting accurate exploratory data analysis.

2. Dataset Overview

The dataset consists of raw weather-related variables collected over multiple timestamps. Initial inspection indicated the presence of missing values, inconsistencies, and numerical features requiring normalization.

	country	location_name	latitude	longitude	timezone	last_updated_epoch	last_updated	temperature_celsius	temperature_fahrenheit	condition_text	...	a
0	Afghanistan	Kabul	34.52	69.18	Asia/Kabul	1715849100	2024-05-16 13:15	26.6	79.8	Partly Cloudy	...	
1	Albania	Tirana	41.33	19.82	Europe/Tirane	1715849100	2024-05-16 10:45	19.0	66.2	Partly cloudy	...	
2	Algeria	Algiers	36.76	3.05	Africa/Algiers	1715849100	2024-05-16 09:45	23.0	73.4	Sunny	...	
3	Andorra	Andorra La Vella	42.50	1.52	Europe/Andorra	1715849100	2024-05-16 10:45	6.3	43.3	Light drizzle	...	
4	Angola	Luanda	-8.84	13.23	Africa/Luanda	1715849100	2024-05-16 09:45	26.0	78.8	Partly cloudy	...	

5 rows × 41 columns

3. Tasks Performed

3.1 Identification of Missing Values and Inconsistencies

- Performed an audit of the dataset to check for NaNs and incomplete entries.
- Identified columns with significant missing values and potential outliers.
- Verified consistency of units and data ranges (e.g., temperature values, humidity %, wind speed).

Checking Null values ouptut

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 106208 entries, 0 to 106207
Data columns (total 41 columns):
 #   Column                        Non-Null Count   Dtype
---  ------                        --------------   -----
 0   country                       106208 non-null  object
 1   location_name                 106208 non-null  object
 2   latitude                      106208 non-null  float64
 3   longitude                     106208 non-null  float64
 4   timezone                      106208 non-null  object
 5   last_updated_epoch            106208 non-null  int64
 6   last_updated                  106208 non-null  object
 7   temperature_celsius           106208 non-null  float64
 8   temperature_fahrenheit        106208 non-null  float64
 9   condition_text                106208 non-null  object
 10  wind_mph                      106208 non-null  float64
 11  wind_kph                      106208 non-null  float64
 12  wind_degree                   106208 non-null  int64
 13  wind_direction                106208 non-null  object
 14  pressure_mb                   106208 non-null  float64
 15  pressure_in                   106208 non-null  float64
 16  precip_mm                     106208 non-null  float64
 17  precip_in                     106208 non-null  float64
 18  humidity                      106208 non-null  int64
 19  cloud                         106208 non-null  int64
...
 39  moon_phase                    106208 non-null  object
 40  moon_illumination             106208 non-null  int64
dtypes: float64(23), int64(7), object(11)
memory usage: 33.2+ MB
```

3.2 Handling Missing Data

- Cleaned the dataset by handling NaN values appropriately.
- Applied imputation strategies such as:
    - Mean/median replacement for numerical features.
    - Forward-fill/backward-fill where sequential continuity was necessary.

This preprocessing ensures the dataset does not contain gaps that could impact downstream analysis.

Replaced NAN values Output

```
           country     location_name  latitude  longitude         timezone  \
0      Afghanistan             Kabul   34.5200    69.1800       Asia/Kabul
1          Albania            Tirana   41.3300    19.8200    Europe/Tirane
2          Algeria           Algiers   36.7600     3.0500   Africa/Algiers
3          Andorra  Andorra La Vella   42.5000     1.5200   Europe/Andorra
4           Angola            Luanda   -8.8400    13.2300    Africa/Luanda
...            ...               ...       ...        ...              ...
106203   Venezuela           Caracas   10.5000   -66.9167  America/Caracas
106204     Vietnam             Hanoi   21.0333   105.8500     Asia/Bangkok
106205       Yemen             Sanaa   15.3547    44.2067        Asia/Aden
106206      Zambia            Lusaka  -15.4167    28.2833    Africa/Lusaka
106207    Zimbabwe            Harare  -17.8178    31.0447    Africa/Harare

        last_updated_epoch      last_updated  temperature_celsius  \
0               1715849100  2024-05-16 13:15                 26.6
1               1715849100  2024-05-16 10:45                 19.0
2               1715849100  2024-05-16 09:45                 23.0
3               1715849100  2024-05-16 10:45                  6.3
4               1715849100  2024-05-16 09:45                 26.0
...                    ...               ...                  ...
106203          1763019000  2025-11-13 03:30                 25.2
106204          1763019000  2025-11-13 14:30                 27.0
106205          1763019000  2025-11-13 10:30                 16.2
106206          1763019000  2025-11-13 09:30                 22.7
106207          1763019000  2025-11-13 09:30                 24.2
...
106206                 43
106207                 43

[106208 rows x 41 columns]
```

3.3 Normalization and Aggregation of Numerical Features

- Applied **Min-Max Scaling** to normalize the numerical variables.
- Transformed all continuous features into a uniform range (0 to 1).
- Ensured that features with different scales (e.g., temperature vs. wind speed) now contribute proportionally in further analysis and modeling.

Normalization Output

	country	location_name	latitude	longitude	timezone	last_updated_epoch	last_updated	temperature_celsius	temperature_fahrenheit	condition_text	...
0	Afghanistan	Kabul	0.719014	0.689521	Asia/Kabul	0.0	NaN	0.695007	0.694153	Partly Cloudy	...
1	Albania	Tirana	0.783594	0.550251	Europe/Tirane	0.0	NaN	0.592443	0.592204	Partly cloudy	...
2	Algeria	Algiers	0.740256	0.502934	Africa/Algiers	0.0	NaN	0.646424	0.646177	Sunny	...
3	Andorra	Andorra La Vella	0.794689	0.498617	Europe/Andorra	0.0	NaN	0.421053	0.420540	Light drizzle	...
4	Angola	Luanda	0.307824	0.531657	Africa/Luanda	0.0	NaN	0.686910	0.686657	Partly cloudy	...

5 rows × 41 columns

Aggregation Output:

	latitude	longitude	last_updated_epoch	temperature_celsius	temperature_fahrenheit	wind_mph	wind_kph	wind_degree	pressure_mb	pressure_in	...	gust_kph
date												
2024-05-31	19.211157	21.571142	1.716479e+09	25.146254	77.262991	8.896344	14.320574	169.975227	1013.077644	29.915350	...	22.333444
2024-06-30	19.183897	21.743390	1.718499e+09	26.458191	79.626374	9.127356	14.694160	178.469251	1011.956417	29.882174	...	22.217657
2024-07-31	19.104503	22.120577	1.721100e+09	26.801201	80.244012	8.693869	13.995249	183.542488	1011.525457	29.869846	...	20.843492
2024-08-31	19.107315	22.088584	1.723811e+09	26.791679	80.226683	8.641522	13.912407	180.549545	1011.988586	29.883474	...	20.411166
2024-09-30	19.103744	22.084018	1.726442e+09	25.124671	77.226911	8.215216	13.224226	169.930415	1013.015216	29.913498	...	18.951906
2024-10-31	19.143867	22.503692	1.729068e+09	22.335382	72.204086	7.602126	12.237375	168.554651	1014.646512	29.962058	...	16.539236
2024-11-30	19.152160	22.633795	1.731712e+09	19.424566	66.966495	7.545711	12.148272	159.128589	1015.511604	29.987767	...	16.986729
2024-12-31	19.107269	22.088991	1.734345e+09	17.751894	63.953780	8.116890	13.065691	163.333333	1016.078246	30.004304	...	18.372787
2025-01-31	19.180727	22.675545	1.737028e+09	17.392724	63.309158	8.078022	13.004829	164.437063	1015.991841	30.001558	...	18.340826
2025-02-28	19.115330	22.085891	1.739571e+09	17.424684	63.366942	8.283013	13.333773	160.083746	1017.145135	30.035684	...	18.434451
2025-03-31	19.110843	22.102421	1.742118e+09	20.095299	68.173432	8.242013	13.267721	161.620427	1014.315180	29.951998	...	18.006042

4. Outcome

After performing the above tasks:

- The dataset is now fully cleaned and free from missing or invalid values.
- All numerical fields are normalized and standardized for analysis.
- The processed dataset is ready for Exploratory Data Analysis (EDA) and model development in subsequent milestones.

5. Conclusion

The preprocessing activities completed in this milestone significantly enhance data quality and usability. With clean, normalized, and consistent data, reliable insights can now be derived in the next phases of the project.